

3rd year

إشارات ونظم معالجة
ذكا ومرتبة ثالثة

Signals & Systems Lab

Lab (2)

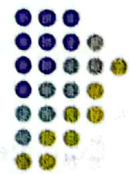
Plotting Several Signals

Lecturer: Amjed B.Jumaah

Ahmed Santa



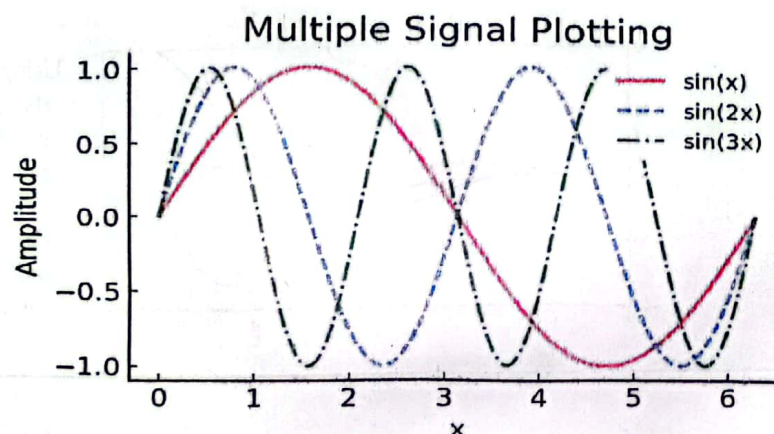
Introduction

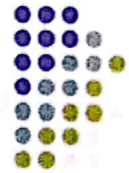


In this lab, we will learn how to plot multiple signals in MATLAB.

We will cover:

- Using linspace to generate the time domain.
- Plotting several functions in the same figure.
- Customizing colors, markers, and line styles.





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Plotting Several Signals

Lecturer: Amjed B.Jumaah

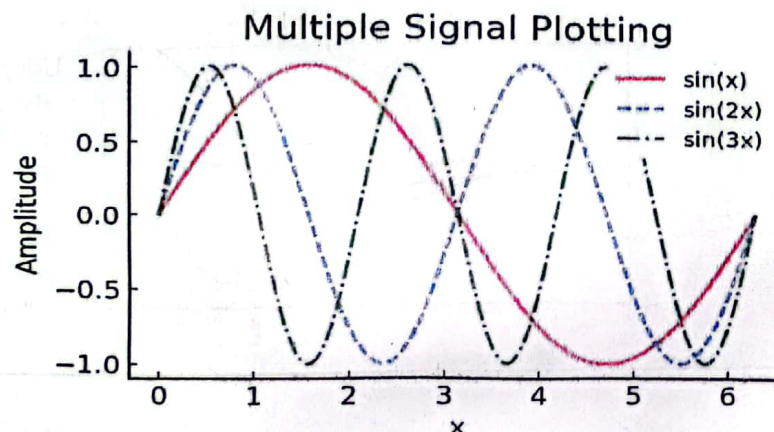
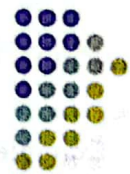


Introduction

In this lab, we will learn how to plot multiple signals in MATLAB.

We will cover:

- Using linspace to generate the time domain.
- Plotting several functions in the same figure.
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linspace

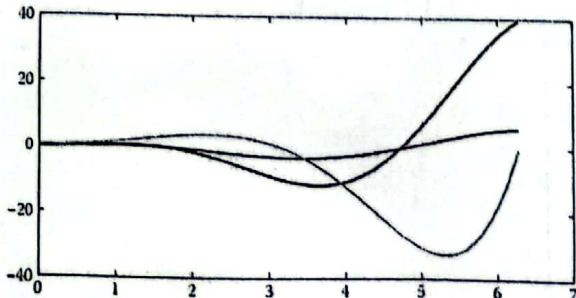
$x = \text{linspace}(\text{initial_value}, \text{final_value}, \text{number_of_elements})$

Commands	Results	Comments
<code>x=linspace(0,1,5)</code>	<code>ans = 0 0.25 0.50 0.75 1.00</code>	Vector x from 0 to 1 with 5 elements.
<code>x=0:1/4:1</code>	<code>ans = 0 0.25 0.50 0.75 1.00</code>	Equivalent vector creation.
<code>logspace(0,1,5)</code>	<code>ans = 1.00 1.78 3.16 5.62 10.00</code>	The command <code>logspace</code> creates 5 logarithmically spaced points on the interval $[10^0, 10^1]$.

Plotting Several Functions in One Figure

Example

Plot the functions $y(x) = x^2 \cos(x)$, $g(x) = x \cos(x)$, and $f(x) = 2^x \sin(x)$, $0 \leq x \leq 2\pi$, in the same figure.

Commands	Results	Comments
<code>x=linspace(0,2*pi,100);</code>		Definition of vector x .
<code>y=(x.^2).*cos(x);</code> <code>g=x.*cos(x);</code> <code>f=(2.^x).*sin(x);</code>		Definition of vectors y , g , and f .
<code>plot(x,y,x,g,x,f)</code>		Using this syntax of the command <code>plot</code> , it is possible to plot many functions in the same figure. Each function is plotted in different color.

Symbols and Styles

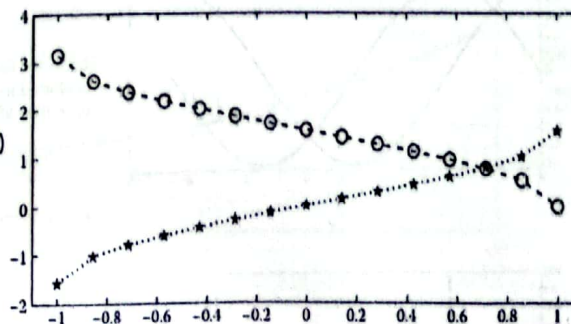
Symbol	Color	Symbol	Point Type	Symbol	Line Type
b	Blue	.	Point	-	Solid
g	Green	o	Circle	:	Dotted
r	Red	x	x-mark	-. ,	Dashdot
c	Cyan	+	Plus	--	Dashed
m	Magenta	*	Star		
y	Yellow	s	Square		
k	Black	d	Diamond		
w	White	<, >	Triangle		
		p	Pentagram		
		h	Hexagram		



Drawing Several Functions with different styles

Commands	Results	Comments
<code>x=linspace(-1,1,15);</code>		Definition of vector x.
<code>y1=asin(x);</code>		Definition of vectors
<code>y2=acos(x);</code>		y1 and y2.

`plot(x,y1,'k:p',x,y2,'r-o')`



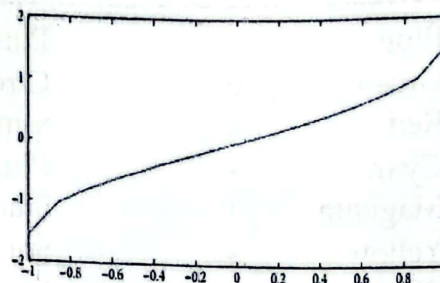
The first function is plotted by black pentagons with dotted line, while the second function is plotted by red circles with dashed line.



Cont.

Commands	Results	Comments
----------	---------	----------

```
plot(x,y1,'Color',[1 0 1], 'Linewidth',4)
```



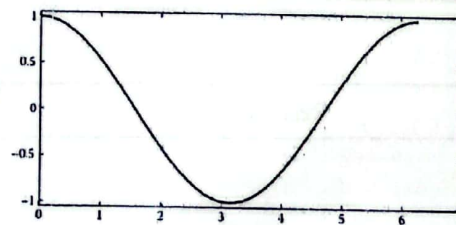
The vector that appears after the text "Color" specifies the RGB elements that take values on [0 1]. Number 4 determines the line width.



Cont.

Commands	Results	Comments
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```
x=linspace(0,2*pi,150);  
plot(x,cos(x));
```

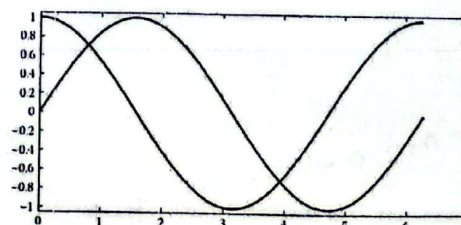


Definition of vector x and direct plot of $\cos(x)$.

hold on

The old graph is preserved.

```
plot(x,sin(x));
```

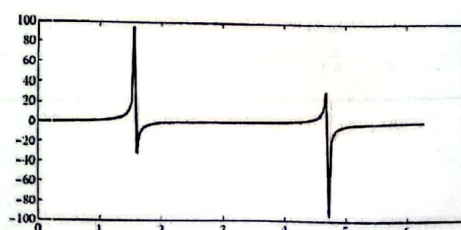


The new function is plotted together with the old one.

hold off

Switch to the non-hold mode.

```
plot(x,tan(x))
```



The old graph is discarded and the new function is plotted alone.



Logarithmic

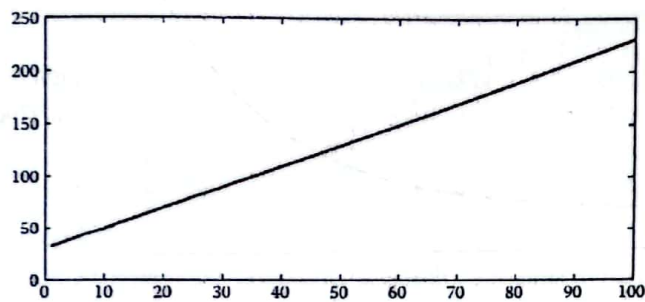
loglog-semilogx-semilogy-area-fplot-ezplot

Commands

Results

Comments

```
x=1:100;  
y=2*x+30;  
plot(x,y)
```



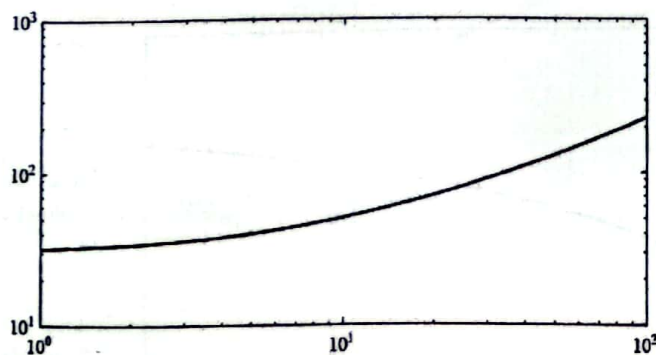
Graph of the function
 $y(x) = 2x + 30, 1 \leq x \leq 100.$



Logarithmic

loglog-semilogx-semilogy-area-fplot-ezplot

```
loglog(x,y)
```

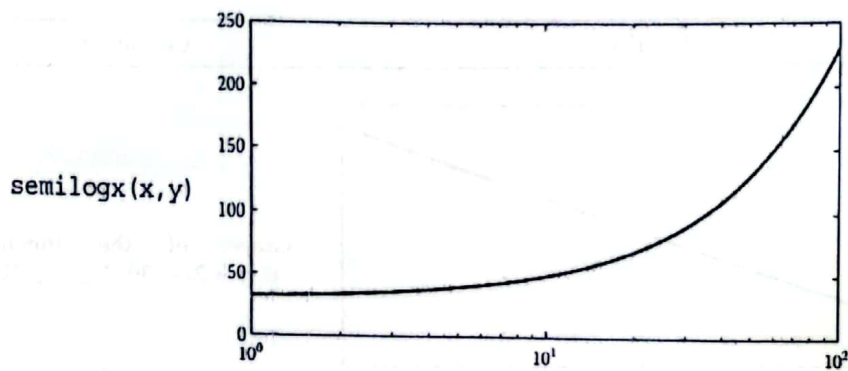


Graph of $y(x)$ in logarithmic
scale axes.



Cont.

loglog-semilogx-semilogy-area-fplot-ezplot

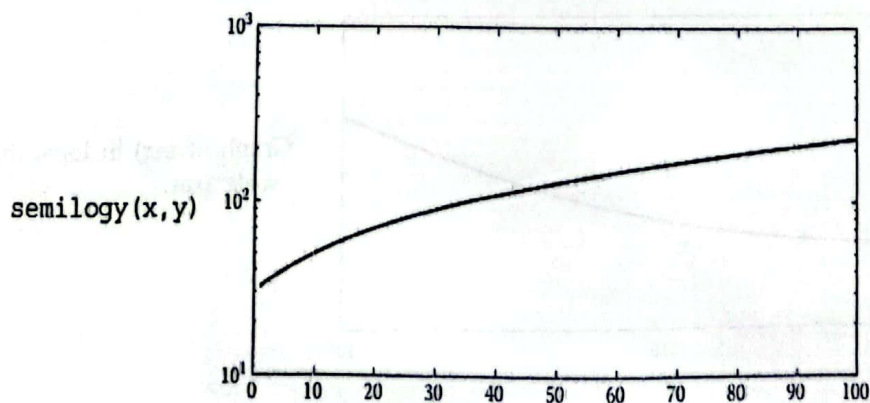


Graph of $y(x)$ with the x -axis being logarithmically scaled.



Cont.

loglog-semilogx-semilogy-area-fplot-ezplot

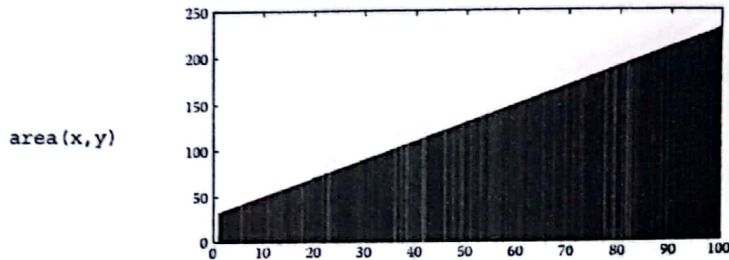


Graph of $y(x)$ with the y -axis being logarithmically scaled.



Area

loglog-semilogx-semilogy-area-fplot-ezplot



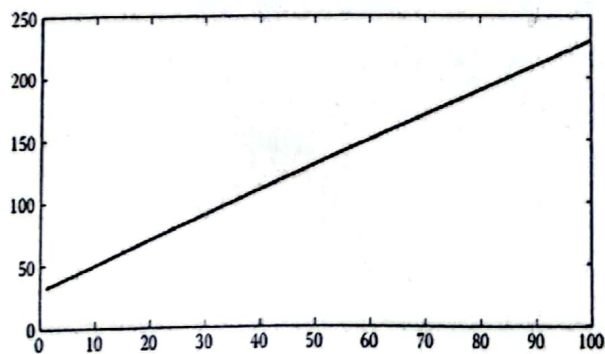
A stacked area plot of $y(x)$.

13

Fast way plotting

loglog-semilogx-semilogy-area-fplot-ezplot

fplot('2*x+30', [1,100])
% or
ezplot('2*x+30', [1,100])



Fast ways of plotting $y(x)$. By using fplot or ezplot there is no need to create vectors.

Assignments

Plot the functions:

$$y_1(x) = \sin(2x), y_2(x) = \sin(3x), y_3(x) = \sin(4x)$$

for $0 \leq x \leq 2\pi$, in the same figure.

Thank you